Generative AI-driven knowledge management in manufacturing firms: a five-stage framework for dynamic knowledge optimization and digital innovation

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Abstract

Purpose – *As artificial intelligence (AI) technologies continue evolving, the transformation of knowledge and information paradigms offers new perspectives on comprehensive innovation in knowledge management (KM) within manufacturing firms. This study aims to explore the innovative application of generative artificial intelligence (GenAI) in the KM of manufacturing firms, to address challenges such as the acquisition of tacit knowledge, cross-departmental silos and dynamic knowledge optimization and to promote the effective use of knowledge resources and intelligent innovation.*

Design/methodology/approach – *This study combines literature analysis with Chinese manufacturing case studies to develop a five-phase GenAI-enhanced KM framework (acquisition, sharing, integration, application and optimization). Through empirical validation, this study establishes an intelligent KM innovation model integrating explicit-tacit knowledge dynamics and GenAI's technical features.*

Findings – *This study constructs scenarios demonstrating how GenAI can facilitate intelligent KM in manufacturing firms. These scenarios broaden the channels for knowledge acquisition, sharing, integration and application, thereby contributing to the development of a logical model and a proposed operational architecture for intelligent KM within such firms.*

Research limitations/implications – *This study has limitations including GenAI implementation costs, data privacy concerns and industry-specific applicability. Future research should address cost-effective implementation, enhanced data privacy measures and cross-sector adaptation.*

Originality/value – *By proposing specific scenarios in which GenAI can be leveraged to enhance intelligent KM, this study refines a logical model and operational architecture that have the potential to significantly improve the efficiency of knowledge utilization. The findings of this study provide practical guidance and theoretical support for manufacturing firms aiming to leverage GenAI to enhance KM and foster innovation.*

Keywords *Generative artificial intelligence, Knowledge management, Manufacturing firms, Digital transformation, Knowledge optimization*

Paper type *Viewpoint*

Received 25 March 2025
Revised 23 July 2025
25 September 2025
Accepted 18 October 2025

Funding: This research was supported by National Key Research and Development Program of China [Grant No. 2024YFC3810802], The National Social Science Fund of China [Grant No. 24AGL016] and The Beijing Municipal Social Science Foundation [Grant No. 22GLB033].

1. Introduction

The transformative advances in generative artificial intelligence (GenAI), particularly exemplified by the advent of large language models such as ChatGPT, are fundamentally reshaping the modalities of human knowledge acquisition, analysis and utilization. As the cornerstone of the national real economy and a critical arena for international industrial competition, the manufacturing industry is strategically vital. Its digital, networked and intelligent transformation, in particular, greatly boosts the sector's overall competitiveness (Liu *et al.*, 2024). The World Economic Forum released a report in 2023 entitled The Impact of the Fourth Industrial Revolution on the Supply Chain [1]. One of its key findings is that manufacturing firms implementing digital, networked and intelligent transformation strategies have achieved significant cost reduction and revenue growth.

In today's landscape of intelligent manufacturing, firms increasingly recognize knowledge as a foundational strategic resource. Although conventional artificial intelligence (AI) finds broad application within knowledge management (KM) systems, its contributions typically remain incremental rather than transformative (Dong *et al.*, 2025). Persistent departmental barriers and organizational rigidity continue to impede the smooth circulation of knowledge. This not only obstructs the sharing of valuable tacit expertise but also reinforces isolated "knowledge islands" across different business units (Toscani, 2023). GenAI, however, presents a fundamentally different opportunity for reshaping KM practices. For knowledge mining, GenAI excels at autonomously producing diverse content formats (text, images, etc.), moving beyond the data-centric analysis and programmed decision rules that characterize traditional AI systems (Wang and Zhang, 2024; Leoni *et al.*, 2024). When it comes to knowledge creation, the technology does not just create content independently but also refines operational parameters. This fosters true collaborative innovation between humans and AI, dramatically boosting innovation output (Benbya *et al.*, 2024; Jarrahi *et al.*, 2023). Regarding knowledge dissemination, the adoption of intuitive natural language interfaces effectively dismantles longstanding barriers to knowledge sharing. This significantly streamlines how knowledge is converted and applied throughout an organization (Liu *et al.*, 2024).

However, investigations into how GenAI intersects with KM are still at a preliminary stage. Current insights remain largely disconnected and lack definitive conclusions. Applying resource orchestration theory, Wang and Zhang (2024) empirically demonstrates GenAI's positive impact on green KM. Yet other researchers contend that this technology may yield complex consequences for organizational KM – under certain circumstances potentially undermining knowledge initiatives altogether (Alavi *et al.*, 2024; Chin *et al.*, 2024). Given these unresolved tensions, our research seeks to comprehensively examine three critical issues: What role does GenAI play in transforming how manufacturing firms manage their knowledge workflows? How can manufacturers leverage GenAI-driven KM frameworks to attain superior performance results? Through what mechanisms does GenAI enhance human–AI partnerships within manufacturing knowledge management processes (KMPs)?

Theoretically, this study systematically constructs the core application scenarios of GenAI in KM within manufacturing firms. By delving into their underlying logic models and operational architectures, this research provides a comprehensive extension to the existing research in this domain. Practically, through a case study of the digital transformation at Zhongtian Group, this research demonstrates the feasibility and effectiveness of the proposed application scenarios and models. The resulting implementation framework offers manufacturers feasible guidance for deploying intelligent knowledge systems, specifically addressing how to optimize human–AI teamwork and strengthen overall KM effectiveness.

The remainder of this paper is organized as follows. Section 2 summarizes and reviews the literature on GenAI and its potential applications in KM. In Section 3, this paper introduces the dilemmas faced by existing KM in manufacturing firms. Building on these insights, Section 4 reconstructs the knowledge logic architecture and application scenarios for manufacturing firms based on GenAI to address these challenges. Section 5 then evaluates the case of Zhongtian's digital transformation and summarizes the operational architecture of KM informed by GenAI. The concluding section provides a summary of the article, highlighting research insights and future research directions related to intelligent KM, particularly concerning organizational change and risk challenges.

2. Theoretical background

2.1 Generative artificial intelligence technology fundamentals and applications

GenAI refers to a class of AI models capable of learning patterns from training data and generating new, multimodal content that resembles the original data. Common generative models include variational autoencoders, generative adversarial networks and autoregressive models. GenAI has already demonstrated significant potential for application across a variety of fields. Within education, GenAI aids student understanding by producing customized learning resources for complex topics. Intelligent tutoring systems support learning outcomes through personalized feedback (Bahroun *et al.*, 2023). In health care, GenAI enhances clinician efficiency and diagnostic precision. By simulating patient cases and assessment tools, it can generate medical history summaries, assist diagnosis and inform treatment strategies (Boscardin *et al.*, 2024). For business operations, GenAI acts as a catalyst for innovation. It fosters novel business models, streamlines processes and elevates customer interactions (Kanbach *et al.*, 2024). Mondal *et al.* (2023) also highlight its critical role in supporting data-driven choices. The technology produces detailed reports and advanced forecasting models, so that organizations can make strategic decisions with greater precision.

GenAI makes use of complicated neural network architectures, particularly deep learning techniques, to achieve outputs comparable to human creations. When it comes to handling visual data, machine vision has brought profound changes to manufacturing through capabilities like automated inspections, enhanced quality assurance and streamlined processes. Within furniture production, this technology is highly capable of tasks such as information gathering, quality checking, precise location and automated sorting (Li *et al.*, 2023). Convolutional neural networks (CNNs) are powerful tools for pattern recognition. They produce impressive outcomes across various application areas, including image analysis, natural language processing (NLP) and medical imaging diagnostics. The integration of machine vision with advanced technologies such as CNNs provides a new technical foundation and innovative ideas for the extraction of tacit knowledge in manufacturing firms.

2.2 Artificial intelligence-enabled knowledge management processes

With the rapid development of AI, many scholars have begun to explore the intersection of AI and KM. Leoni *et al.* (2024) emphasize that integrating AI into KMPs not only accelerates the pace of knowledge dissemination but also enhances organizations' decision-making processes. Iaia *et al.* (2024) constructed a human-process-technology model and used AI to address the deficiencies in knowledge, identification, skill development and introduction in firm innovation. In addition, according to the stimulus-organization-response theory, Tian *et al.* (2023) verified, through empirical tests, that AI has a significant positive impact on the innovation performance of firms' green technology development by promoting basic, complementary and extended knowledge coupling. In conclusion, the introduction of an intellectual KM system can create a broader development space for manufacturing firms.

Regarding knowledge acquisition specifically, AI can significantly enhance the efficiency of knowledge generation through automated data collection and preprocessing (Taherdoost and Madanchian, 2023). Cross-disciplinary research by Jarrahi *et al.* (2023) reveals how contemporary deep learning advances have dramatically enhanced algorithmic capacities to emulate human cognitive functions – particularly in visual recognition, speech interpretation, natural language understanding and analytical reasoning. These developments establish critical foundations for intelligent KM systems. Balaram *et al.* (2023) analyzed Huawei Technologies' marketing strategies to develop a media ecology-AI framework. This approach leverages sensor technology and nodal analysis for real-time market monitoring, substantially enhancing data accuracy. In the realm of knowledge application, Leoni *et al.* (2022) demonstrate how AI implementation in manufacturing boosts supply chain resilience and organizational performance through KMPs. A survey they conducted empirically on 120 Italian manufacturers confirmed this positive relationship between AI and both KMP effectiveness and business outcomes. Ottersböck *et al.* (2024) addressed the critical challenge of capturing retiring experts' tacit knowledge by conceptualizing, designing and deploying an AI-powered autonomous learning support system for inexperienced workers. For knowledge conversion, Nonaka and Takeuchi's (1995) seminal SECI model outlines four conversion modes between tacit and explicit knowledge. Complementing this framework, complementary to this, the theory of "ba" (Nonaka and Konno, 1998) posits that knowledge transformation occurs within specific physical or virtual spaces, necessitating deliberate creation of enabling environments for effective knowledge generation. Building on these foundations, Cerchione *et al.* (2024) proposed the WISED model – an evolution of SECI principles adapted for contemporary data-driven contexts. This refined framework unlocks novel pathways for organizational knowledge development and conversion processes.

Although the application of GenAI in firm KM has gradually increased in recent years, existing research primarily focuses on the significance of AI for KM and its role in the KMPs of firms (Huang *et al.*, 2025). However, it does not delve into the specific implementation of GenAI within the context of firm KM intelligence. In the era of the digital economy, there is an urgent need for digital transformation in manufacturing firms. In this study, we innovatively conceptualize the application scenarios for KM in manufacturing settings, seek to organically coordinate GenAI with the KMPs and integrate GenAI into the KMPs and offer KM strategies based on GenAI to facilitate the digital transformation of manufacturing firms.

3. The realistic dilemma facing knowledge management in manufacturing firms

KM is the process of creating, acquiring and applying knowledge to enhance organizational performance (Bassi, 1997). Modern manufacturing firms typically use systematic processes to integrate knowledge resources. However, the management focus is predominantly on the acquisition and application of explicit knowledge (Samtani *et al.*, 2023; Liu *et al.*, 2024). Tacit knowledge, such as employees' operational experience and skills, remains difficult to effectively capture and share (Ottersböck *et al.*, 2024). In particular, against the backdrop of exponential data growth, the limitations of traditional management models have become increasingly apparent (Taherdoost and Madanchian, 2023):

- Manufacturing firms face challenges in knowledge acquisition because of complex production scenarios, diverse equipment and varied personnel roles. These conditions typically generate fragmented information streams resistant to systematic integration. Firms also struggle to extract and use tacit knowledge, hindering the conversion of knowledge assets into growth drivers. In 2023, 87.9% of global firms prioritized data and analytics investment, according to Statista [2]. When companies lack timely operational insights, firms inevitably face compromised decision quality and diminished productivity relative to market peers.
- The knowledge-sharing mechanism has significant shortcomings. In the knowledge economy era, departments often operate in silos, pursuing their own unique goals and priorities. This siloed mode of operation leads to frequent information barriers, low efficiency and the prevalence of knowledge silos (Sleep *et al.*, 2023). For instance, Nokia's failure to transition to smartphones effectively, because of departmental barriers and knowledge silos, led to the forced sale of its mobile phone business. The stagnation of knowledge sharing not only weakens decision-makers' ability to swiftly capture market dynamics and respond but also restricts their vision in formulating forward-looking decisions and plans.

- Manufacturing firms encounter substantial knowledge integration challenges arising from diverse operational workflows, multifaceted data streams and varied knowledge representations. Production involves multiple departments and complex operations, leading to fragmented and hard-to-manage data and knowledge. Digital transformation initiatives further intensify these difficulties by demanding consolidation of heterogeneous data from IoT sensors, ERP platforms and MES environments. For example, Tesla has faced increased production complexity and delivery inefficiencies because of challenges in integrating data from IoT and ERP systems.
- Knowledge application has limitations. Tacit knowledge, including the skills and experiences of employees within a firm, is not adequately converted into shareable knowledge resources. This inadequacy restricts the dissemination and utilization of knowledge (Huang *et al.*, 2022). Compounding this issue, inadequate knowledge transformation systems coupled with insufficiently targeted training and guidance hinder employees from transferring learned concepts to real-world challenges. Because of suppliers' failure to convert tacit knowledge of pedal design into shared standards, Toyota suffered from safety hazards and brand trust damage.
- Knowledge optimization is often overlooked. Many firms become preoccupied with immediate knowledge capture and short-term use, overlooking the essential practice of continual enhancement. This absence of effective knowledge renewal mechanisms allows knowledge repositories to stagnate, rendering them progressively misaligned with evolving market realities and technological advancements. Moreover, the rapid evolution of knowledge and accelerated shifts in concepts pose significant challenges for employees to adapt.

In light of the above analysis, current KM practices are insufficient in enhancing corporate competitiveness. Rather than prioritizing archival functions like knowledge consolidation and storage, contemporary KM must evolve beyond static repositories to focus on actionable intelligence that drives organizational synergy and unlocks strategic value. Against the backdrop of an increasingly digital and interconnected society, it is imperative to break down internal information silos and develop a more flexible and intelligent KM system (Manlio Del Giudice and Armando Papa, 2023). This evolving context makes digital transformation essential for modern firms. Knowledge digitization has become a key strategy for achieving efficient KM.

4. Generative artificial intelligence-enabled knowledge management innovation logic architecture for manufacturing firms

To foster the development and innovation of manufacturing firms, it is essential to enhance the mining and application of tacit knowledge. Based on algorithms and computational power, AI offers a promising solution in this regard. According to a research report by Dune Intelligence [3], the application of GenAI in firms is on the rise. An analysis of 276 cases reveals that GenAI has been implemented across 32 application scenarios within organizations, as illustrated in Table 1. The report indicates that functions such as knowledge assistance, intelligent customer service and data analysis are prevalent in most firms, serving as key drivers of intelligent transformation in KM. The expanding diversity of GenAI applications suggests substantial potential for pioneering new KM paradigms.

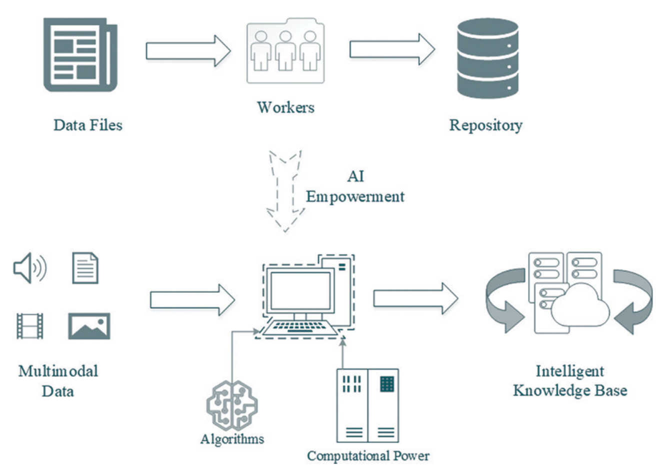
In the era of KM intelligence, the traditional information transformation pathway of data–employee–knowledge-base is evolving into a more advanced, multidimensional pathway of multimodal-data–AI-center–intelligent-knowledge-base, as illustrated in Figure 1. During the transformation of data into knowledge, key processing tasks can be effectively handled by AI, which demonstrates a low error rate when processing complex information and enhances both the efficiency and accuracy of the knowledge transformation process. The established intelligent knowledge base can interact with

Table 1 Distribution of generative artificial intelligence application scenarios

Application scenarios	%
Knowledge assistant	13.0
Intelligent customer service	12.7
Data analysis	9.8
Employee office assistant	8.3
Coding assistant	6.5
Marketing content generation	4.7
Intelligent investment research	4.3
Intelligent document assistant	4.0
Product design	3.6
Intelligent investment adviser	3.3
Composite data	2.9
Virtual assistant	2.9
Drug discovery	2.5
Intelligent operation and maintenance	2.2
Intelligent cockpit	1.8
Credit approval	1.8
Software testing	1.8
Sales assistant	1.8
Audio and video content generation	1.8
Fault diagnosis and prediction	1.8
Personalized experience	1.4
Fraud detection	1.1
Trading robot	1.1
Manufacturing process optimization	1.1
Fund management	0.7
Underwriting claims	0.7
Risk early warning	0.4
Video parsing	0.4
Material discovery	0.4
Intelligent account opening	0.4
Product dynamic pricing	0.4
Chip design	0.4

Source(s): Compiled by authors from the case materials of GenAI of the Dune Intelligence

Figure 1 Transformation of information pathways



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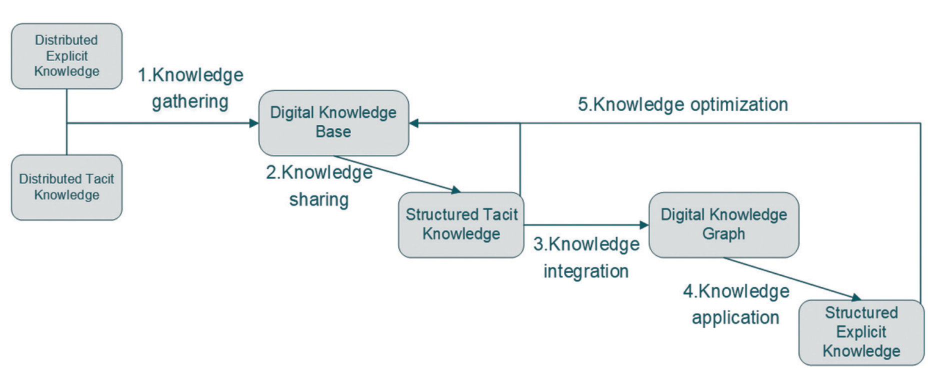
employees through natural language models, efficiently store and retrieve information and continuously optimize its content through personalized recommendations. By learning and analyzing employees' query patterns, it can also provide instant decision support for users.

This paper presents five application scenarios for KM intelligence in manufacturing firms, leveraging the SECI KM transformation model (Nonaka and Takeuchi, 1995) and the WISED (Cerchione *et al.*, 2024) model of knowledge transformation in the data-driven era. The integration of GenAI enhances the entire KMPs in manufacturing, encompassing collection, dissemination, integration, application and optimization. This approach ultimately facilitates the transformation of manufacturing firms from an "experience-driven" to a "knowledge-driven + AI-driven" innovation paradigm, as illustrated in Figure 2.

4.1 Intelligent knowledge collection: gathering dispersed explicit and tacit knowledge to build a digital knowledge base

The intelligent collection and management of knowledge is essential for firms to effectively facilitate knowledge sharing and application. Through GenAI, manufacturing firms can efficiently establish digital knowledge bases, achieve the rapid digitization and standardized management of both explicit and tacit knowledge and prevent the loss and misalignment of knowledge resources. First, the collection and organization of explicit knowledge serve as foundation for the intelligent acquisition of knowledge. Manufacturing firms generate a substantial number of document files, including manuals, design drawings and other materials, throughout the production process. This knowledge, which can be articulated and communicated in formal, systematic language, is referred to as explicit knowledge (Nonaka and Takeuchi, 1995). Typically, these materials are stored in corporate management information systems or databases by employees according to specific classification criteria. Accessing this information often demands comprehensive analysis and the examination of multiple related databases or information systems. Besides, manufacturing firms derive substantial value from diverse external intelligence sources. Regulatory updates accessed via government portals, shifting market indicators and corporate disclosures from industry peers all represent vital inputs for developing comprehensive knowledge repositories. It involves burdensome collection and storage processes to manage vast volumes of internal and external firm data. Besides, it often requires refinement before supporting decisions or operations. GenAI significantly alleviates this challenge. Specifically, GenAI systematically gathers industry reports and patent documents through its multimodal data capture capabilities, effectively digitizing explicit knowledge. Building on this functionality, NLP facilitates advanced semantic annotation. This approach actively identifies essential technical parameters – such as material formulations and manufacturing

Figure 2 Knowledge intelligence management



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specifications – within complex documents. This process generates standardized knowledge components, establishing groundwork for subsequent sharing and application.

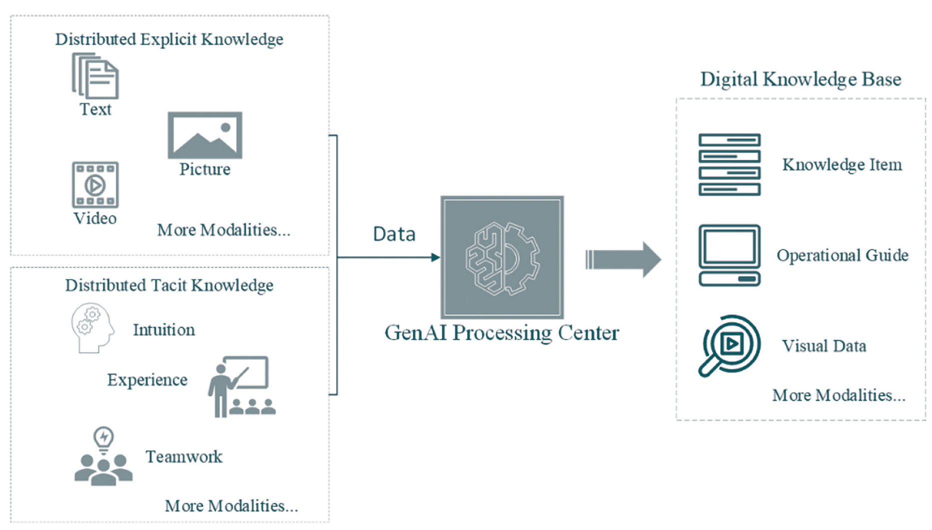
Moreover, extracting tacit knowledge represents the pivotal stage in intelligent knowledge acquisition. Tacit knowledge often defies straightforward articulation via traditional communication methods. Nevertheless, its considerable strategic value renders this knowledge indispensable for driving organizational innovation and sustaining competitive positioning. Tacit knowledge is embodied in the personal abilities of employees and reflected in their operational skills, experience, intuition and judgment. Traditional models may introduce biases regarding the externalization of tacit knowledge, making it challenging for future generations to understand and apply this knowledge. By using machine vision, CNNs and other image-processing technologies, GenAI converts records of employees' operational processes into visual data. The technology identifies critical techniques by interpreting physical movements and verbal communications during work processes, generating visual operation guides. Concurrently, speech recognition algorithms convert expert dialogues into searchable digital knowledge, as illustrated in Figure 3, effectively codifying previously tacit knowledge.

4.2 *Intelligent knowledge sharing: using a digital knowledge base to enhance the accumulation of structured tacit knowledge*

Knowledge is a prerequisite for employee innovation, and the act of sharing knowledge is the path to achieving employee innovation (Jiang and Chen, 2018). Firms can establish a knowledge-sharing cloud platform by using GenAI's extensive mining of the digital knowledge base. GenAI-driven intelligent knowledge sharing is no longer a one-way transmission. Rather, it has evolved into a dynamic, interactive process that accelerates collaboration and innovation among employees. This approach facilitates sustained growth and value creation while generating systematic and structured tacit knowledge resources.

First, GenAI develops a personalized knowledge recommendation system by deeply analyzing employees' behavioral data, interest preferences and role-specific requirements. It can accurately identify the learning needs of employees in various positions. For example, process

Figure 3 Knowledge intelligence gathering



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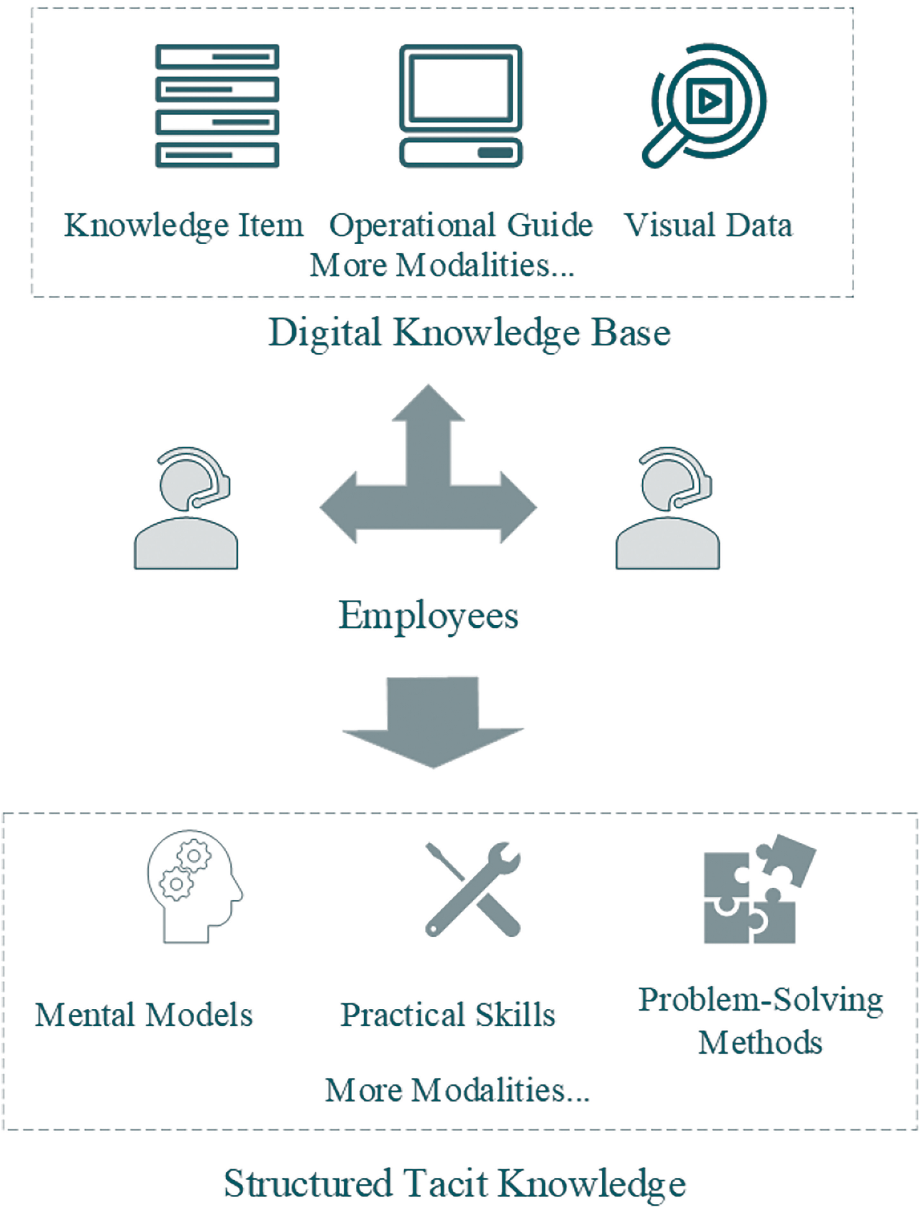
engineers may prioritize optimization and adjustment methods for process parameters, while production line operators require guidelines for equipment operation and troubleshooting skills. By delivering role-aligned learning content, GenAI ensures professionals access immediately applicable knowledge. Employees rapidly locate relevant information and implement new skills directly in their workflows. This process concurrently strengthens operational fluency and inventive capacity, catalyzing organizational resilience and continuous development trajectories.

In addition, with the help of GenAI, firms can create a borderless knowledge-sharing space, as illustrated in Figure 4. Human–AI interaction transforms knowledge exchange from unidirectional transfer to bidirectional co-creation, enhancing cognitive synergy between

Figure 4 Intelligent knowledge sharing



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human expertise and algorithmic processing (Zhao *et al.*, 2023). In this space, employees can upload their immediate ideas. AI automatically optimizes the processes of collecting, organizing and distributing knowledge through algorithmic models. Through continuous cross-employee dialogue, the platform progressively refines knowledge content quality. Crucially, such systems integrate cross-departmental knowledge flows across organizational hierarchies. Through intelligent analysis and data mining, the system can identify potential connections between pieces of knowledge and foster innovative thinking. Machine learning enables continuous evolution of knowledge-sharing platforms. Such recommendation mechanism can ensure the timely update and accuracy of knowledge. The platform systematically documents and analyzes operational data and experiential insights of the employees, generating proprietary organizational knowledge resources. Meanwhile, the borderless knowledge sharing space cuts meeting times and training costs significantly, while narrowing interdepartmental information gaps. This approach effectively eradicates corporate knowledge silos and strengthens organizational responsiveness in dynamic markets.

4.3 Intelligent knowledge integration: Structuring tacit knowledge into digital knowledge graphs

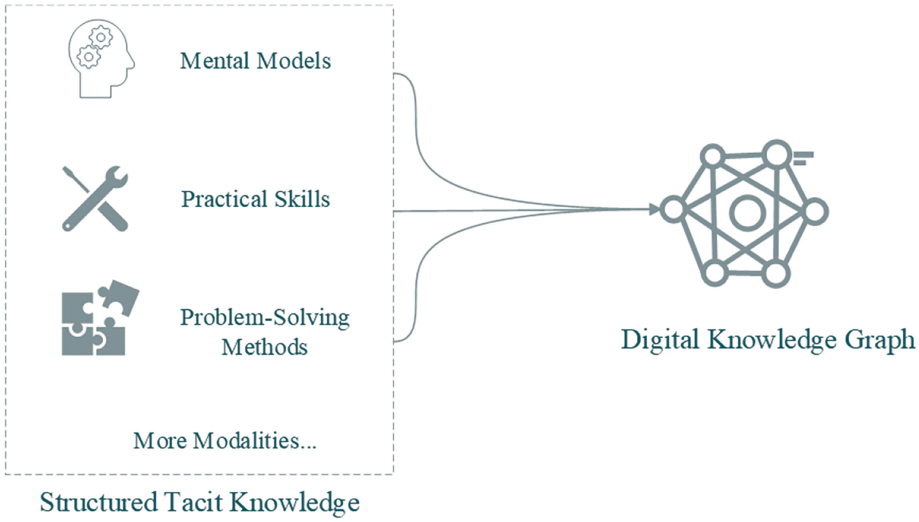
In the third phase, GenAI integrates structured tacit knowledge from Phase 2 to develop organizational knowledge graphs. First, GenAI uses its modeling capabilities to map out relationships between core entities, building an accurate ontology framework in the process. These frameworks comprehensively reflect the characteristics of structured tacit knowledge established earlier, which in turn ensures the accuracy and completeness of graphs. Second, GenAI uses its data preprocessing capabilities to standardize structured implicit data, removing duplicate, invalid or erroneous data. This establishes data consistency and comparability. The standardized data will be integrated with the existing firm data (such as ERP systems, CRM systems, etc.) to form a more comprehensive knowledge base. In the knowledge extraction and integration stage, GenAI first aligns the structured implicit data formed in the second stage with the ontology model and extracts detailed information. It then uses NLP technology to supplement knowledge from unstructured data. This technology focuses on the correlation between data in the two stages, thereby constructing a unified knowledge graph. The verification mechanism of GenAI ensures the accuracy of knowledge and provides a guarantee for the quality of the knowledge graphs.

After constructing the knowledge graph, operators rely on GenAI’s graph database management function to choose an appropriate graph database for storage. Meanwhile, GenAI’s visualization tool presents the knowledge graph in an intuitive graphical form, which enhances knowledge readability and comprehensibility. To maintain the timeliness and accuracy of the knowledge graph, the real-time data update function of GenAI can promptly capture and incorporate new or modified data. At the same time, optimization algorithms use application feedback to continuously improve extraction, fusion and modeling processes. This persistent refinement keeps the system aligned with organizational needs and delivers robust support for intelligent management systems. This is illustrated in Figure 5.

4.4 Knowledge intelligence application: value realization of knowledge graph to facilitate structured explicit knowledge accumulation

Manufacturing firms use GenAI to deploy intelligent applications of digital knowledge graphs. These applications integrate several key tools: intelligent search, decision-making, Q&A and risk anticipation systems. The intelligent search module makes full use of the knowledge graph’s structured information. By combining it with GenAI’s intelligent search algorithm, it provides employees with a fast and accurate information retrieval experience. Employees only need to input their search needs in natural language. The system then quickly locates and returns relevant information from the knowledge graph, which boosts

Figure 5 Knowledge intelligence applications



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work efficiency. Contextual recommendation engines also serve up potentially valuable information using user query histories and behavioral analytics.

Second, intelligent decision support systems are an important part of the application of knowledge intelligence. Knowledge graphs provide multi-dimensional data connections, which empowers firms to drive analysis and decision-making with more comprehensive data (Deng *et al.*, 2025). GenAI uses data mining and machine learning algorithms to uncover key business insights and trend projections from these graphs. These insights, in turn, inform strategic resource allocation. During specific decision-making processes, GenAI can create virtual simulation environments to simulate various decision scenarios and outcomes. This helps decision-makers intuitively assess the risks and benefits of different options. At the same time, the virtual simulation space can mitigate erroneous knowledge caused by GenAI's opaque mechanisms and prevent knowledge contamination (Wach *et al.*, 2023).

Third, the intelligent Q&A module relies on GenAI's NLP capabilities and the vast knowledge stored in knowledge graphs to build conversational interfaces. When employees ask questions with natural language, the system can interpret the semantics of their queries and pull precise responses from the graphs. This kind of conversational interaction greatly eases information acquisition and fosters knowledge exchange within the organization.

Finally, the intelligent risk anticipation systems are an important guarantee for the intelligent application of knowledge. These systems leverage GenAI's predictive algorithms to proactively detect operational vulnerabilities when analyzing graph correlations and data flows. In customer relationship management, these systems identify transactional anomalies or behavioral deviations and send preemptive alerts about potential fraud risks. Once risks are identified, the system can automatically generate response plans or suggestions. These help the firm take timely prevention and control measures, reducing potential losses.

GenAI's computational capabilities and analytical algorithms facilitate deep mining and correlation of firm data, yielding high-value explicit knowledge artifacts. These encompass recommended insights, decision-support business intelligence and precise Q&A responses. Accumulating and disseminating such knowledge enhances operational efficacy while expanding corporate knowledge graphs, establishing foundations for sustained innovation. This is illustrated in Figure 6.

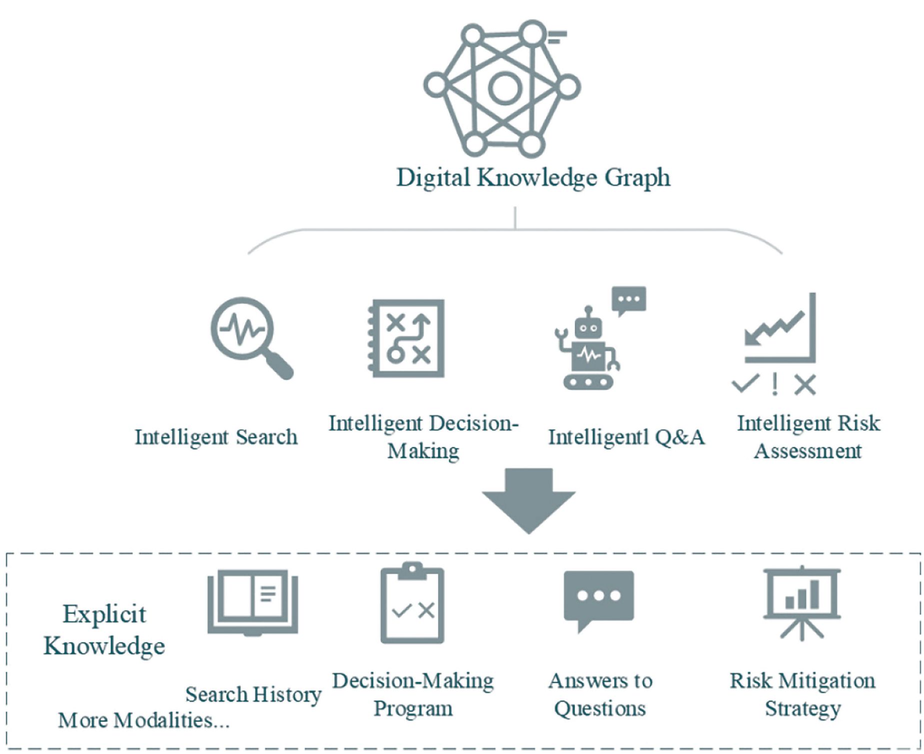
4.5 Knowledge intelligence optimization: cycling and upgrading of knowledge management processes

As pivotal KM tools, digital knowledge graphs deliver precise and reliable knowledge support (Deng *et al.*, 2025). Their core information requires continuous evolution to reflect shifting business demands and external conditions. GenAI-driven mechanisms, thus, sustain autonomous knowledge graph refinement and progressive iteration.

GenAI’s NLP and textual analysis capacities conduct periodic knowledge graph scans, identifying potential inaccuracies or inconsistencies including spelling errors, grammatical flaws and logical conflicts. Detected anomalies undergo automated or human-assisted correction. Integrated real-time data processing continuously harvests emerging knowledge from internal systems and external sources (e.g. industry reports and research publications), dynamically updating knowledge graphs. Concurrently, GenAI prioritizes content based on the timeliness and significance of the knowledge. Deep learning and graph neural networks of GenAI further mine entities and relationships to infer novel associations and patterns, progressively enriching knowledge graphs.

Feedback mechanisms integrate knowledge graph optimization with collection processes, establishing iterative refinement cycles. Each cycle incorporates explicit and tacit knowledge outputs from prior iterations as new collection targets, enabling continuous knowledge enhancement. This methodology progressively refines knowledge systems

Figure 6 Knowledge intelligence applications



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toward greater precision and utility within manufacturing contexts, significantly amplifying organizational value creation.

5. Case validation

5.1 Case selection

In case selection, this study follows the principle of typicality (Yin, 2017). Zhongtian Steel Group (Huai'an) New Materials Co., Ltd. is engaged in the deep processing of high-end metallic materials, specializing in steel wire rope and tire bead wire. Commissioned on January 8, 2022 with a total investment of RMB20.2bn, the company is establishing a plant with an annual capacity of 1.6 million tons of premium steel wire rope. Upon full operation, this facility is expected to supply approximately 30% of global wire-rope output and over 50% of domestic output.

Central to its operations is an end-to-end intelligent-manufacturing system that positions the facility as the first "lighthouse factory" in the metallic-materials deep-processing sector. By embedding information technologies, intelligent algorithms, large-scale industrial deployments, NLP and AI analytics throughout the discrete steel-rope manufacturing process, the plant has achieved comprehensive intelligence from equipment level to managerial decision-making. Consequently, the case provides a highly intelligent and representative context in which the paper empirically examines the applicability and robustness of the proposed framework.

Primary data collection encompassed archival analysis of Zhongtian Corporation's digital assets, including corporate portals across subsidiaries, functional units and affiliated entities. Systematic retrieval covered academic publications, patent filings, media reports and relevant monographs, supplemented by Web-based news coverage of the firm's digital transformation initiatives. Semi-structured interviews were subsequently conducted virtually with key personnel from KM, R&D and standardization departments, with all sessions documented comprehensively.

5.2 Case description

Steel-cord production is traditionally fragmented, with ageing plants, low intelligence and chaotic shop-floors that complicate knowledge extraction. Zhongtian counters these constraints by deploying smart workshops: intelligent equipment, precise material routing and real-time process control integrate previously siloed systems and data. An intelligent scheduler, an energy-and-carbon management system and big-data analytics now generate high-fidelity data streams that underpin conversational business intelligence and accelerate firm-level digital transformation. Through an in-depth exploration of intelligent algorithmic models, large model industrial applications, NLP and AI analysis, Zhongtian has integrated GenAI, a cutting-edge information technology tool, into the KMPs of steel cord manufacturing:

1. First is digital knowledge base construction. Zhongtian leverages GenAI to systematically acquire and encode explicit and tacit knowledge. NLP-based expert systems automatically categorize and annotate textual corpora, while computer vision extracts tacit knowledge from manufacturing contexts. Through coordinated 5G networks, digital twin implementations and virtual reality (VR)/augmented reality (AR) infrastructures, real-time workshop monitoring and immersive visualization become operational. Conversational business intelligence powered by GenAI transforms manufacturing process data into operational intelligence. After algorithm optimization, GenAI applies historical price signals to adjust the operating speed of the equipment. This approach can reduce the peak fluctuations in power demand, while improving asset use and cost-effectiveness.

2. Second is digital knowledge repository implementation. Digital knowledge repositories facilitate boundaryless knowledge exchange through dedicated, centralized cloud platforms. With unified configuration management and real-time data sharing, connectivity could stretch across internal departments, external devices and supply chain partners. This architecture breaks down information silos and systematizes tacit knowledge resources into structured repositories.
3. Zhongtian uses graph database technology to store and manage digital knowledge graph data. This approach not only handles complex relational data but also accelerates query performance and real-time updates, creating durable KM infrastructure. The company has also created visualization interfaces that let employees explore information intuitively. These interfaces produce comprehensible network diagrams of complex knowledge architectures, which clarify semantic connections and hierarchical arrangements to strengthen user understanding.
4. To get the most value from its knowledge graph, Zhongtian developed an intelligent search system that uses graph structures. This tool makes it easy to quickly find relevant knowledge through keyword searches. Unlike traditional search tools, this engine relies on the networked relationships between entities and attributes in the knowledge architecture. It interprets the search context and delivers ranked responses, which significantly improves how relevant and useful the results are. Zhongtian has also built predictive frameworks using intelligence derived from the graph. These models allow for evidence-based forecasting of potential outcomes across different decision scenarios, supporting organizations in making strategically informed choices through comprehensive risk–benefit analysis.
5. Zhongtian strengthens its digital knowledge graphs using streaming data processing frameworks. These frameworks collect insights from diverse real-time sources like news feeds, social media streams and sensor outputs. Then they integrate them directly into knowledge architectures. This kind of integration boosts the temporal relevance of the graphs and improves their accuracy. The company also adopted the incremental update method. This method only modifies the data changes in the knowledge graph. It can maintain the efficiency of resource use and ensure the timeliness of data.

Zhongtian's GenAI-driven KM has cut workforce costs significantly, while also supporting organizational restructuring and efficiency gains. These improvements have made Zhongtian an industry benchmark for intelligent transformation in steel cord manufacturing. With its smart manufacturing systems now fully deployed, the group expects potential revenue to hit RMB50bn – a 40% increase from its previous performance.

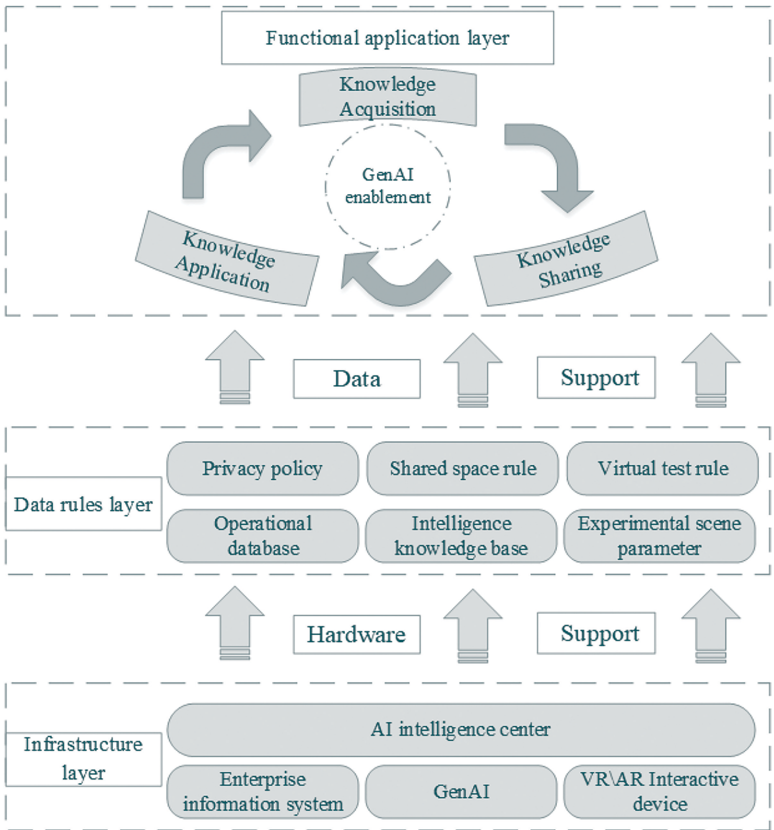
5.3 Operational architecture for generative artificial intelligence-based knowledge management

The implementation of intelligent KM practices in the manufacturing industry needs a solid technological foundation, an effective governance structure and applicable application frameworks. As shown in Figure 7, this operational architecture relies on three key elements: infrastructure, data rules and functional applications.

5.3.1 The infrastructure layer. Intelligent KM systems rely fundamentally on AI, along with VR and AR technologies. It requires strong AI capabilities to build digital knowledge bases and semantic networks within unified system frameworks. At the same time, VR and AR foster immersive environments that support contextual learning. In firm systems, GenAI turns organizational data into operational insights. By integrating multiple intelligent interaction devices, the AI center can provide a solid framework for firm KM.

5.3.2 The data rules layer. For manufacturing firms to achieve comprehensive innovation KM, they need the support of corresponding software, data and rules. This includes

Figure 7 Operational architecture



Source: Authors' own work

databases, knowledge bases and scenario parameters that support various application spaces. In addition, the operators also need to establish rules to manage interactions within the knowledge-sharing community and regulate operational behaviors in virtual experimentation scenarios. These data and rules ensure that intelligent KM is carried out in an orderly manner.

5.3.3 The functional application layer. With support from the infrastructure layer and the data rule layer, the functional application realizes various KM functions and applications. The intelligent knowledge center collects scattered explicit and tacit knowledge in a digital repository, establishing a platform where employees can collaborate with one another. The AI center creates a digital knowledge graph that facilitates both knowledge absorption and its application. Intelligent KM drives comprehensive firm innovation by gradually developing additional functional modules.

This operational architecture gathers and restructures informational resources. Then, it can internalize organizational expertise. Employees subsequently generate novel knowledge assets by existing foundations. Collective intelligence circulates via firm knowledge communities, forming organizational wisdom repositories that drive innovation. Resulting operational data re-enter the knowledge cycle, refining established management processes. As Figure 7 illustrates, this enables intelligent processing of explicit knowledge with real-time dissemination, effective extraction of implicit operational insights and validation within virtual environments – collectively enabling progressive knowledge augmentation.

6. Conclusions and implications

6.1 Conclusions

The competitiveness of manufacturing firms fundamentally depends on the effectiveness of their KM. Drawing on seminal research (Nonaka and Takeuchi, 1995; Cerchione *et al.*, 2024), this study focuses on the five core elements of KM. By taking advantages of GenAI, this study demonstrates its transformative potential to enable intelligent knowledge acquisition, sharing, integration, application and optimization. Furthermore, through the development of a five-stage logic model and a corresponding operational architecture, this study offers manufacturing firms a comprehensive, innovation-driven, AI-enabled KM solution:

1. *Dual-path intelligent knowledge acquisition mechanism*:
   - The structured pathway for explicit knowledge uses NLP techniques to convert technical documents and process manuals into standardized knowledge entries.
   - The digital conversion pathway for tacit knowledge uses machine vision to analyze worker actions and speech recognition to transcribe experience-based narratives. This synergistic framework systematically transforms operative skills and judgmental heuristics into digitized work instructions, establishing a unified structured digital knowledge repository.
2. *Intelligent knowledge sharing model*: First, a personalized recommendation mechanism is constructed from employee behavioral data and job demands to precisely align knowledge supply. Concurrently, a boundary-less sharing space is established, where human–machine collaboration streamlines knowledge-flow processes. It enables real-time, cross-hierarchical interaction and algorithm-driven content optimization. Finally, breaking down departmental barriers creates meaningful knowledge exchange and collaborative innovation.
3. *Self-evolving digital knowledge graph*: Firms can build a domain ontology grounded in core entities like employees, projects and products, which acts as a unified framework for knowledge organization. On this basis, GenAI carries out cleaning, standardization and integration on multi-source data. Then it integrates structured and unstructured data to extract interrelated knowledge. It ensures knowledge accuracy via a calibration mechanism, thus facilitating the autonomous iteration of the knowledge graph.
4. *Knowledge application ecosystem*: Based on knowledge graph, firms can develop an intelligent search system. Simultaneously, an intelligent decision-support system is constructed. An intelligent Q&A module is developed, enabling natural-language interaction coupled with knowledge-graph integration. Additionally, an intelligent risk-early-warning mechanism is established to identify latent risks and generate countermeasures. This yields a closed innovation loop of "application–precipitation–re-application" while continuously sedimenting structured explicit knowledge.
5. *Self-iterative knowledge graph mechanism*: An intelligent verification module is devised to periodically detect subjective inconsistencies and logical contradictions within the graph, enabling precise rectification. Leveraging real-time data-crawling capabilities, endogenous and exogenous knowledge sources are integrated to dynamically update graph content and recalibrate temporal weights. A bidirectional feedback mechanism is instituted, wherein explicit and tacit knowledge sedimented from the preceding cycle is re-injected as new inputs, thereby sustaining an iterative optimization loop.

6.2 Theoretical implications

This study makes two primary theoretical contributions:

First, this study enriches research on the integration of KM and AI by proposing a Smart Innovation Methodology. Existing literature predominantly examines the impact of traditional AI technologies on overall KM effectiveness (Zawaideh and Bataineh, 2025; Jarrahi *et al.*, 2023; Schryen *et al.*, 2025), yet largely neglects in-depth analyses of micro-level mechanisms and application patterns within specific KM processes (Zhang *et al.*, 2025; Iaia *et al.*, 2024). In contrast, although the potential of GenAI in KM is gaining attention (Zu *et al.*, 2025; Rezaei, 2025; Alavi *et al.*, 2024; Chin *et al.*, 2024), research remains nascent, producing fragmented insights and revealing a critical empirical gap: the absence of systematic validation regarding how its core functionalities (e.g. tacit knowledge extraction and content generation) are embedded within KM practices to generate value.

To address these gaps, this study advocates an integrated approach combining technological and managerial innovation. This study delineates precise deployment pathways for GenAI across core KMPs, substantiated through an empirical case analysis. This systematically validates the method's efficacy in enhancing knowledge workflows, filling critical voids in systematic empirical research on AI-driven KM. Thus, it provides novel theoretical frameworks and empirical foundations in this domain.

Second, this research draws on conventional KM frameworks to construct a logic model that fits Industry 4.0 needs. Traditional models function in settings characterized by limited knowledge scale and human-centered interactions (Wang *et al.*, 2025; Jarrahi *et al.*, 2023), with their main focus on how individuals absorb, transform and disseminate knowledge (Krause, 2024; Huang *et al.*, 2025; Ottersböck *et al.*, 2024). Their core paradigm centers on human-driven knowledge flows. With GenAI driving it, the Industry 4.0 era is marked by three key knowledge characteristics: exponentially growing volume, increasing complexity and dynamic fluidity. In this situation, AI moves beyond being a supportive tool to become an active participant in knowledge creation, acquisition, processing and application. Close integration of machine learning, data mining and human–computer collaboration has made human–machine symbiosis the leading paradigm in modern KM.

In this transformative background, this paper assesses the applicability limitations of traditional frameworks and propose a novel logic model for KM. This integrated approach harmonizes human cognition with AI systems, placing collaborative human–AI partnerships at the center. It prioritizes complementary dynamics across knowledge discovery, codification, sharing, application and innovation cycles. GenAI serves as the fundamental enablers for intelligent KM through its analytical and content creation capabilities. As a result, this framework not only fits contemporary industrial knowledge environments but also establishes novel theoretical constructs for sustainable knowledge architectures in the AI era.

6.3 Practical implications

This study's developed KM framework holds substantial practical value for manufacturing firms. It shows how GenAI can be implemented in specific scenarios and builds AI-focused operational structures for knowledge systems, offering actionable guidance for adoption in industry:

1. For employees, intelligent KM makes it easier to acquire knowledge and apply it to daily work. This enhances their work efficiency in return. Furthermore, AI's introduction has broken the constraints of traditional organizational structures on knowledge sharing. It boosts information exchange among employees and strengthens cross-organizational collaboration across the firm. These changes effectively eliminate knowledge silos.

2. For firms, an innovative KM model speeds up knowledge circulation and expands both the breadth and depth of knowledge acquisition. It also strengthens their ability to transform knowledge into practical value. These advancements help manufacturing firms gain a firm foothold in today's highly competitive market.
3. The evolution of KM logic fosters organizational change. The combination of AI, knowledge platforms and decision-making brings to light the urgent need for agile organizational structures and workflows. Traditional hierarchical may put off knowledge dissemination and slow down decision-making (Leoni et al., 2024). To enhance firm and workforce compatibility with AI-enhanced knowledge frameworks, organizations must implement the following structural modifications:
   - *Cultural and mindset change*: Firms need to cultivate a data culture and an innovative consciousness. Firms need to build a data-centric cultural atmosphere from the top down, so that employees can understand and accept the value of data. They should integrate data into daily decision-making and work processes and learn to support decisions with data and facilitate innovation with AI. Senior managers must first have a firm belief and profound understanding of digital transformation. To do this, they need to lead by example for employees through their own actions. Only when organizations develop AI-aligned cultural paradigms can GenAI-driven KM achieve sustainable implementation.
   - *Structural and process reorganization*: Firms need to adapt to the rapidly changing market environment and realize efficient information flow. To do this, they should actively adopt a flat organizational structure and build a networked organizational model. AI-driven knowledge sharing requires a more flexible organizational structure and quicker information response capabilities. This means organizations must simplify their hierarchical structure and standardize work processes. The networked structure grants employee greater independence. What is more, its closer integration with GenAI helps further drive knowledge sharing and resource flow across the organization.

However, AI-generated outputs carry inherent bias and accuracy risks, requiring integration with expert decision systems. Human validation mechanisms filter and rectify contextually discordant information, mitigating decontextualization pitfalls while enhancing knowledge fidelity through systematic knowledge audits.

6.4 Limitations and future directions

KM based on GenAI not only holds significant advantages in firm innovation but also faces numerous challenges. This study identifies three core limitations. First is cost issues. If manufacturing firms hope to achieve comprehensive industrial applications of AI, then they may need to invest substantial R&D funds in the initial stage (Wach *et al.*, 2023). Second, privacy issues are urgent to be solved. Although GenAI injects new vitality into KM, related privacy protection issues must be taken seriously. AI must strictly comply with relevant regulations when processing sensitive data (Samtani *et al.*, 2023; Preiksaitis and Rose, 2023); otherwise, it may trigger serious legal and ethical issues. In addition, there are intellectual property issues: Does the knowledge generated by GenAI have intellectual property rights? In addition, does GenAI infringe on others' intellectual property rights and copyrights?

Future research directions may include the following:

- Innovative solutions are explored to address cost challenges. This study examines the cost challenges faced by large manufacturing firms, while also acknowledging that SMEs encounter significant capital expenditures, staff training costs and other risks when implementing intelligent construction practices. It is essential to find a balance between capital investment and anticipated benefits. Additionally, strategies for overcoming barriers related to personnel and technological changes, as well as those for effectively manage resistance to change, warrant further investigation.

- Research on a KM framework centered on data encryption is essential. When firms use GenAI to gather external data, they must also prioritize the protection of individuals' private information to prevent significant privacy violations.
- It is important to explore the application of the model in service industries, such as health care and education. As compared to the approach adopted above, the integration of GenAI into other industries can be explored in the future based on more quantitative data and a wider range of survey samples.

Notes

[1.] Data source: The Impact of the Fourth Industrial Revolution on the Supply Chain, https://cn.weforum.org/publications/

[2.] Data source: Statista, Global state of data/analytics investment 2023 | Statista.

[3.] Data source: Dune Intelligence. www.shaqiu.cn/zhiku

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